

Practice Problems

A 45.0° flint glass prism has a refractive index of 1.6705 for sodium yellow light, and it is adjusted for minimum deviation. Find (a) the angle of minimum deviation and (b) the angle of incidence. (c) Solve graphically.

Two thin prisms have powers of 6.0 D each. At what angles should their axes be superimposed to produce powers of 2.0, 4.0, 6.0, 8.0, 10.0, and 12.0 D?

A direct-vision prism is to be made of two elements like the one shown in Fig. 2K. The flint-glass prism of index 1.720 has an angle $\alpha'' = 55.0^\circ$. Find the angle α' for the crown-glass prism if its refractive index is 1.520. Solve by (a) graphical methods and (b) calculation.

A ray of light is incident on a piece of glass at an angle of 45.0° . If the angle of refraction is 25.37° , find (a) the refractive index and (b) the critical angle. (c) Solve (b) graphically